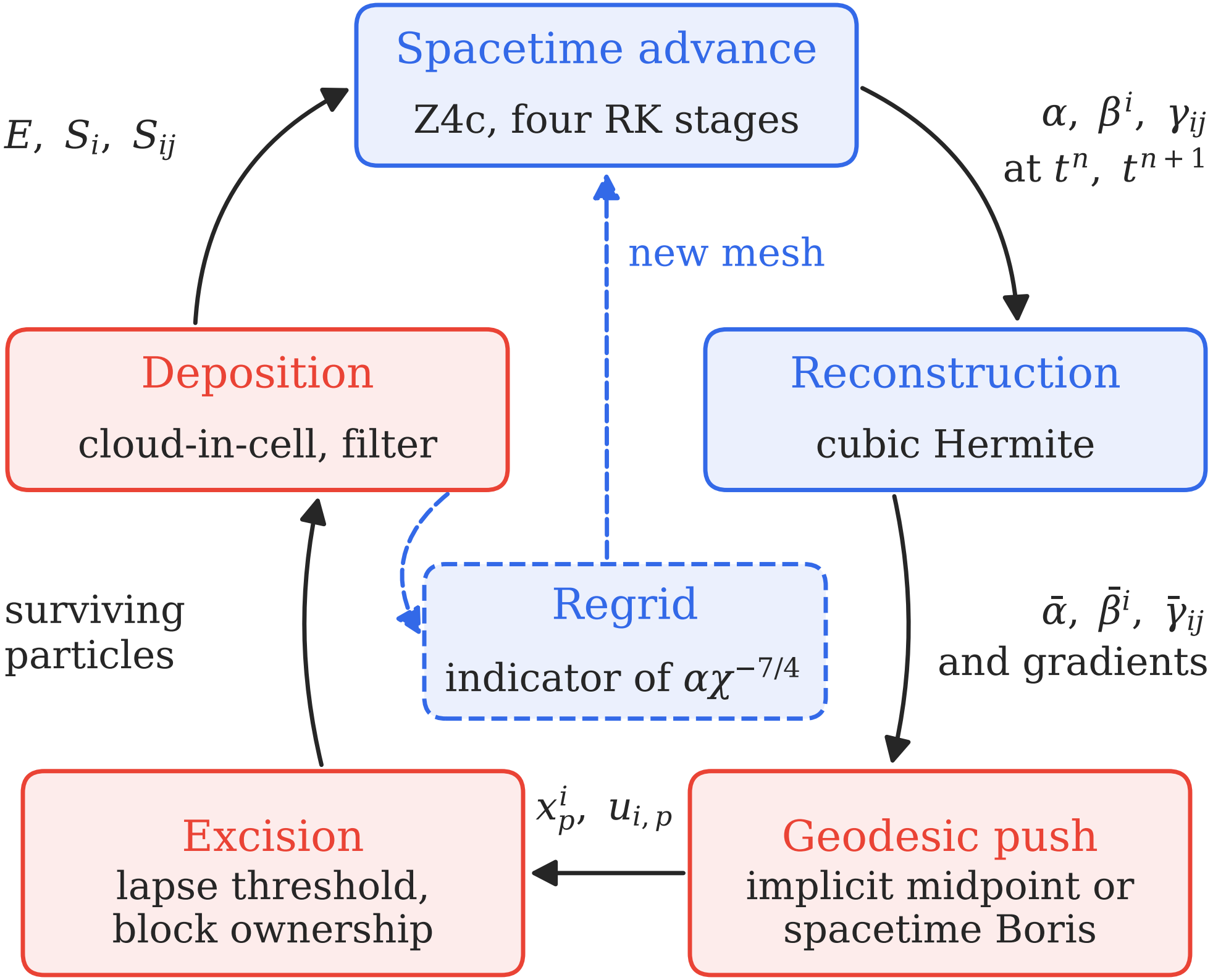
Spacetime advance
Z4c, four RK stages
$E,\ S_i,\ S_{ij}$
$\alpha,\ \beta^i,\ \gamma_{ij}$
at $t^n,\ t^{n+1}$
new mesh
Deposition
cloud-in-cell, filter
Reconstruction
cubic Hermite
surviving
particles
Regrid
indicator of $\alpha\chi^{-7/4}$
$\bar{\alpha},\ \bar{\beta}^i,\ \bar{\gamma}_{ij}$
and gradients
Excision
lapse threshold,
block ownership
$x^i_p,\ u_{i,p}$
Geodesic push
implicit midpoint or
spacetime Boris